Star Plucking: Monotone Lower Bounds for k-List Orthogonal Vectors at Logarithmic Dimension

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Work in progress: not verified

This report records work in progress; it is not peer-reviewed.

- **AI-generated:** proofs, computations and code were produced by Claude (Anthropic, Claude Opus 5.5) in sessions directed by the author, who has not independently verified the mathematics.
- **No human review:** the main proofs were checked only by AI referee agents of the same model family (`docs/ai-review-log.md`).
- **No formal verification:** nothing is formalized in Lean; SAT results have no DRAT certificates.

Everything here still needs thorough verification. Do not cite these results as established.

Abstract

For every fixed $k \geq 2$ and $\varepsilon > 0$ we show that, for all large n, every monotone circuit deciding whether k lists of n Boolean vectors of dimension $d \geq c \log n$ have no orthogonal k-tuple needs $n^{k-\varepsilon}$ gates, where $c = O_k(\varepsilon^{-k})$. For $k = 2$ this is a theorem of Choudhury, Limaye, Sreenivasaiah and Srinivasan [3]. For $k \geq 3$ their bounds need larger, superlogarithmic dimension. The new ingredient, *star plucking*, controls approximator conjuncts that share a core vector. It is applied recursively over *frames*, each a core together with part of a link. We first give the case $k = 3$ in full. We also record notes on routes to refuting the Orthogonal Vectors Hypothesis (OVH), motivated by the claimed refutation of ETH [1, 2]: a conditional algorithm for block-CNF with constant block size, a heuristic no-go for linear deformation designs, and exact equality ranks of DISJ_k for $k \leq 4$. None of this proves or disproves OVH.

1 Introduction

The Orthogonal Vectors problem $\mathrm{OV}_{n,d}$ asks whether two lists of n vectors in $\{0,1\}^d$ contain an orthogonal pair. The *Orthogonal Vectors Hypothesis* (OVH) states that for every $\varepsilon > 0$ there is c such that $\mathrm{OV}_{n,c\log n}$ has no $O(n^{2-\varepsilon})$ algorithm. Its k-list version (k-OVH) asks for $n^{k-\varepsilon}$. OVH was mainly supported by SETH, which implies it [4]. Two October 2026 preprints [1, 2] claim $2^{o(n)}$ algorithms for 3-SAT. If correct, they refute ETH and SETH but not OVH, since their running times degrade with clause width, and OVH loses its main evidence. This raises two questions: which routes could refute OVH, and in which restricted models can it be proved? Theorem 1 below does not depend on these preprints.

For lists $U_1, \ldots, U_k$ of n vectors in $\{0,1\}^d$, let $k\mathrm{Int}_{n,d}(U_1, \ldots, U_k) = 1$ iff $u_1 \wedge \cdots \wedge u_k \neq 0^d$ for every $(u_1, \ldots, u_k) \in U_1 \times \cdots \times U_k$. This is the negation of k-OV, and it is monotone. A monotone circuit has fan-in-2 AND and OR gates; its inputs are the knd input bits and the constants 0 and 1. Its size is its number of gates.

Theorem 1 (main)**.** *For every fixed $k \geq 2$ and $\varepsilon \in (0,1)$ there is $c \leq (k+1)(33(k+1)^2/\varepsilon)^k = O_k(\varepsilon^{-k})$ such that, for all large n and all $d \geq c \log n$, every monotone circuit computing $k\mathrm{Int}_{n,d}$ has at least $n^{k-\varepsilon}$ gates. In particular, monotone circuits for $k\mathrm{Int}_{n,d}$ need $n^{k-o(1)}$ gates for every $d = \omega(\log n)$.*

Relation to prior work. For $k = 2$, Theorem 1 is [3, Thm 5]. For $k \geq 3$, [3] reduce from clique. This gives near-tight monotone bounds at $d = n^{\varepsilon}$ [3, Thm 4] and, for infinitely many n, at $d = \exp((\log n)^{1/2+o(1)}) = n^{o(1)}$ [3, Cor 13]. They explain why that framework cannot reach all $d = \omega(\log n)$, even with optimal clique bounds [3, Remark 16]. To our knowledge the logarithmic-dimension case $k \geq 3$ is not covered by earlier work. In non-monotone models, Lin [9] proves $n^{\Omega(k)}$ lower bounds for k-OV in AC^0 of every fixed depth, for large fixed k, at dimension $O(k \log n)$. That result is incomparable with Theorem 1, which is monotone with the tight exponent. Our literature search after July 2026 has been limited.

Contributions.

1. **Main result (§3).** Theorem 1, proved by recursive star plucking over frames.
2. **The case $k = 3$ (§2),** with explicit constants (Theorem 2). It introduces the method in its simplest form.
3. **Notes on refutation routes (§4):**
   - a conditional algorithm for block-CNF with constant block size (Theorem 14);
   - a heuristic no-go for linear deformation designs (§4.2);
   - exact equality ranks with an LP barrier (Theorem 15, Propositions 16 and 17).

Scope. Monotone lower bounds do not transfer to general algorithms [11, 13]. The main theorem only shows that refuting k-list OVH requires non-monotone steps such as cancellation.

Changes from version 1.

- New Theorem 1 for every fixed k (§3). Version 1's main theorem is now Theorem 2.
- The description of [3] is corrected. Version 1 said that for $k \geq 3$ they reach only $d = n^{\Omega(1)}$ and that they leave the logarithmic case open. In fact their Corollary 13 reaches $d = n^{o(1)}$ for infinitely many n, and they do not pose the logarithmic case as an open problem.
- Version 1's corollary on unbalanced OV (n versus n^2 vectors) is withdrawn. A direct run of the two-list argument of [3] with unequal list sizes very likely gives it, so we do not claim it.
- In §4: Theorem 14 is restated for decision, with its hypothesis derived from the claimed refutation of ETH via [8], and unproved remarks after it are removed. The deformation no-go is demoted to a heuristic. The bound $r(5) \leq 5$ and the earlier 0/1 searches are attributed to [5], explicit certificates for the upper bounds are added, and the #OV record is updated to [7].
- Exposition: all choices in the construction are fixed deterministic rules, the reveal order of Lemma 13 rests on a stated lemma (Lemma 12), related work [9] is added, and notation clashes are removed.

2 The case of three lists

For lists U, V, Z of n vectors in $\{0,1\}^d$, $3\mathrm{Int}_{n,d}(U,V,Z) = 1$ iff $u \wedge v \wedge z \neq 0^d$ for all $(u,v,z) \in U \times V \times Z$.

Theorem 2. *For every $\varepsilon \in (0,1)$ there is $c = O(\varepsilon^{-3})$ such that, for all large n and all $d \geq c \log n$, every monotone circuit computing $3\mathrm{Int}_{n,d}$ has at least $n^{3-\varepsilon}$ gates.*

2.1 Overview

We use Razborov's approximation method [10, 12] in the form of [3]. Each gate is replaced by an *approximator*: a conjunction of monotone pieces, each reading a few vectors. Errors are measured on a distribution $\mathcal{D}_0$ of 0-inputs and a distribution $\mathcal{D}_1$ of 1-inputs. For two lists, CLSS use pieces that each read one vector. For three lists, one-vector pieces lose a factor of only n^{-2} at OR gates, which caps the bound at about n^2, so pieces must read two vectors. This creates a new failure mode, *stars*: many pieces $B_i(u, v_i)$ sharing a core u. When two stars with the same core meet at an OR gate, $\Theta(n^2)$ three-vector pieces must be dropped, at a $\mathcal{D}_0$-cost of $\Theta(1/n)$ per gate. *Star plucking* (conditions (A3)–(A4)) fixes this.

- For each value w of the core, petals that rarely fail under $\mathcal{D}_1$ are switched off. This is cheap by Lemma 4.
- A one-vector check $Q_u(w)$, "few frequently-failing petals", is added. This is cheap on $\mathcal{D}_1$, because many such petals make the star almost surely 0 anyway.

After plucking, the petal that matters on a $\mathcal{D}_0$ sample is a bad one only with probability $\approx 1/n$. That restores an $n^{-3+o(1)}$ error per gate.

2.2 Distributions

$\mathcal{D}_1 = \mu_p$, i.e. each bit is 1 independently with probability p. By a union bound, $\Pr_{\mathcal{D}_1}[3\mathrm{Int} = 0] \leq n^3(1-p^3)^d \leq \frac{1}{2}$ once $d \geq (3 \ln n + 1)/p^3$.

$\mathcal{D}_0$: pick indices $a_U, a_V, a_Z \in [n]$ uniformly and call the vectors with these indices *chosen*. For each coordinate j independently, pick $L_j \in \{U, V, Z\}$ uniformly. The chosen vector of list L_j gets 0 at j, the other two chosen vectors get 1, and all unchosen vectors are 1^d. Every sample is a 0-input.

Lemma 3. *(i) The indices are independent of the contents.*

(ii) Given the indices, each chosen vector has law $\mu_{2/3}$.

(iii) Given the indices, any two chosen vectors jointly dominate $\mu_{1/2}$ on their $2d$ bits.

(iv) Given the indices and a chosen x, another chosen y is a product distribution with $y_j = 1$ where $x_j = 0$, and $\Pr[y_j = 1] = \frac{1}{2}$ where $x_j = 1$.

Proof. (i), (ii) and (iv) follow from the construction; (ii) and (iv) are products with marginals $\geq \frac{1}{2}$. For (iii), couple one coordinate. The pair is uniform on $\{(1,1),(1,0),(0,1)\}$. Map the points $(1,1),(1,0),(0,1)$ of $\mu_{1/2}^{\otimes 2}$ to themselves, and split $(0,0)$ into three masses of $\frac{1}{12}$, one to each point. □

Lemma 4 ([3, Lemma 22])**.** *For monotone B and $0 < p < q < 1$, $\Pr_{\mu_q}[B = 0] \leq \Pr_{\mu_p}[B = 0]^{\lfloor q/p \rfloor}$. Hence, with $t := \lfloor 1/(2p) \rfloor$, every ν dominating $\mu_{1/2}$ satisfies $\Pr_{\nu}[B = 0] \leq \Pr_{\mu_p}[B = 0]^t$.*

2.3 Approximators and star plucking

A *slot* is a set J of one or two vectors from distinct lists. A *conjunct on J* is a monotone function B_J of the bits of J that is *saturated*: $B_J = 1$ whenever some vector of J equals 1^d. Property **(P)** means $\Pr_{\mathcal{D}_1}[B_J = 0] \geq \delta$. Under $\mathcal{D}_0$, a saturated conjunct can be 0 only if all of its vectors are chosen. In particular, for conjuncts B_J and $B'_{J'}$ of two approximators, $B_J \vee B'_{J'} \equiv 1$ on $\mathcal{D}_0$ whenever $J \cup J'$ repeats a list.

Definition 5. An *approximator* is a constant, or $F = \bigwedge_{J \in \mathcal{J}} B_J$ (one conjunct per slot) with a set T of fewer than $2M$ *cores*, such that:

(A1) every conjunct satisfies (P);

(A2) every slot meets T, and each 2-slot J is assigned the core core(J): the vector of $J \cap T$ from the first of its lists in the order U, V, Z. The 2-slots assigned to u are the *petals* of u; their other vectors are distinct.

(A3) *(plucked)* for a petal $J = \{u, x\}$ let $\beta_J(w) := \Pr_{x \sim \mu_p}[B_J(w, x) = 0]$, which is nonincreasing in w. Then $\beta_J(w) \in \{0\} \cup (\eta, 1]$ for all w.

(A4) with $Q_u(w) := [\#\{J : \beta_J(w) > \eta\} < \theta/\eta]$ (monotone, $Q_u(1^d) = 1$): either $B_{\{u\}} \le Q_u$ (*enforced*) or $\Pr_{\mu_p}[Q_u = 0] < \delta$ (*virtual*).

Normalization $N(G)$ of a conjunction G of saturated conjuncts:

1. Drop conjuncts violating (P).
2. Take the maximal family $\mathcal{M}$ of conjuncts with disjoint slots chosen greedily in a fixed order of the slots. If $|\mathcal{M}| \ge M$, output 0. Otherwise set $T = \bigcup \mathcal{M}$ and assign cores.
3. *Pluck*: replace each petal B_J by $B_J \vee [\beta_J(u) \le \eta]$. Then, if $\Pr_{\mu_p}[Q_u = 0] \ge \delta$, AND Q_u into $B_{\{u\}}$, creating $B_{\{u\}} := Q_u$ if that conjunct is absent (also if it was dropped in step 1); otherwise Q_u is virtual.
4. Drop petals that now violate (P). This only raises Q_u, so (A4) is preserved.

Gates.

- Leaves are their own approximators.
- Constant children are handled exactly: $\mathrm{AND}(0, \cdot) = 0$, $\mathrm{AND}(1, F) = F$, $\mathrm{OR}(1, \cdot) = 1$, $\mathrm{OR}(0, F) = F$.
- AND: $N(F_g \wedge F_h)$, merging equal slots.
- OR: for $F_g = \bigwedge B_J$ and $F_h = \bigwedge B'_{J'}$, $N(G)$ with $G = \bigwedge (B_J \vee B'_{J'})$ over the pairs for which $J \cup J'$ is a slot, so $G \ge F_g \vee F_h$.

2.4 Error analysis

At a gate with output A, the $\mathcal{D}_0$-error is $[A = 1, F_g \circ F_h = 0]$ and the $\mathcal{D}_1$-error is $[A = 0, F_g \circ F_h = 1]$.

Lemma 6 (normalization)**.** *If $M \le n$ and G has $\le 4M$ 1-slots and $\le 8Mn + 4M^2$ 2-slots (true at every gate), then* $\Pr_{\mathcal{D}_0}[N(G) = 1, G = 0] \le (40M\delta^t + 4M\eta^t)/n$ *and* $\Pr_{\mathcal{D}_1}[N(G) = 0, G = 1] \le (1 - \delta)^M + 2Me^{-\theta}$.

Proof. Drops. A dropped conjunct is 0 on $\mathcal{D}_0$ only if its slot is chosen (probability $n^{-|J|}$), and then with probability $\le \delta^t$ (Lemmas 3(ii)/(iii) and 4). In total this is $\le 20M\delta^t/n$.

Switch-off. A change needs u, x chosen and $B_J(u, x) = 0$ with $\beta_J(u) \le \eta$. Given u this has probability $\le \eta^t$ (Lemmas 3(iv) and 4). Over fewer than $2M$ cores with $\le 2n$ petals each, this is $\le 4M\eta^t/n$.

Truncation. The members of $\mathcal{M}$ are independent under $\mathcal{D}_1$, each failing with probability $\ge \delta$.

Enforcing Q_u. Given u with $Q_u(u) = 0$, at least θ/η petals with distinct, independent partner vectors each fail with probability $> \eta$. So all of them equal 1 with probability $\le e^{-\theta}$. □

Lemma 7 (OR gate)**.** $\Pr_{\mathcal{D}_0}[G = 1, F_g = F_h = 0] \le 4M\delta^t/n + 26M^2(\theta/\eta)^2/n^3$.

Proof. Let $\mathcal{V}$ be the event that some chosen core with a *virtual* Q_u has $Q_u(u) = 0$. Then $\Pr[\mathcal{V}] \le 4M\delta^t/n$. Work outside $\mathcal{V}$, on a sample with $G = 1$ and $F_g = F_h = 0$.

Failing conjuncts lie inside the chosen triple C. If failing J (of F_g) and J' (of F_h) had $J \cup J' \subsetneq C$, then $B_J \vee B'_{J'}$ would be a failing conjunct of G. So every failing pair spans C.

If a side is a petal of a core u with $Q_u(u) = 0$, then Q_u is enforced, so $B_{\{u\}}$ fails too, and replacing that side by $\{u\}$ yields another failing pair, which must also span. Two single vectors cannot span, so we end with either (a) a single vector against a petal whose core passes Q, or (b) two petals whose cores pass Q.

A failing petal whose core passes Q is moderate ($\beta > \eta$), and fewer than θ/η petals per core value are moderate. By Lemma 3(i), the chosen vector of another list is one of them with probability $< \theta/(\eta n)$.

Two spanning 2-slots share exactly one vector. Counting the configurations:

- (a): $\le 2(2M)^2$ configurations, each with probability $\le n^{-2}\theta/(\eta n)$;
- (b), same core on both sides: $< 2M$ configurations, each $\le n^{-1}(\theta/(\eta n))^2$;
- (b), the core of one side is a petal of the other: $\le 2(2M)^2$, each $\le n^{-2}\theta/(\eta n)$;
- (b), a common petal: $\le (2M)^2$, each $\le n^{-2}\theta/(\eta n)$.

The total is $\le 26M^2(\theta/\eta)^2/n^3$. □

Proof of Theorem 2. Let Φ be a monotone circuit for $3\mathrm{Int}_{n,d}$ with $|\Phi|$ gates and output approximator F. By induction over the gates, $F \ge \Phi$ outside the union of the $\mathcal{D}_1$-errors, and $F \le \Phi$ outside the union of the $\mathcal{D}_0$-errors.

- If $F \equiv 0$, then $\frac{1}{2} \le |\Phi|[(1-\delta)^M + 2Me^{-\theta}]$.
- Otherwise every conjunct meets T, so $\Pr_{\mathcal{D}_0}[F = 0] < 2M/n \le \frac{1}{2}$. Hence $\frac{1}{2} \le |\Phi|\varepsilon_0$ with $\varepsilon_0 = (44M\delta^t + 4M\eta^t)/n + 26M^2(\theta/\eta)^2/n^3$.

Take $t = \lceil 64/\varepsilon \rceil$, $p = 1/(2t)$, $M = n^{\varepsilon/8}$, $\theta = n^{\varepsilon/16}$, $\delta = n^{-\varepsilon/16}$ and $\eta = n^{-4/t}$. Then $\delta^t, \eta^t \le n^{-4}$ and $\theta/\eta \le n^{\varepsilon/8}$, so $\varepsilon_0 \le 27n^{-3+\varepsilon/2}$ and $(1-\delta)^M \le e^{-n^{\varepsilon/16}}$. Therefore $|\Phi| \ge \min\{e^{n^{\varepsilon/16}}/(6n^{\varepsilon/8}), n^{3-\varepsilon/2}/54\} \ge n^{3-\varepsilon}$ for large n, with $c = 4(2t)^3 = O(\varepsilon^{-3})$. □

Remark. For two lists every 2-slot spans, only 1-slots survive, and the argument reduces essentially to [3]. (One difference: truncation here uses a maximal disjoint family and also runs at OR gates.) For $k \ge 4$, conjuncts read up to $k-1$ vectors. Moderate links of a core can then share vectors, so they are no longer independent under $\mathcal{D}_1$. Section 3 plucks recursively to handle this.

3 Every fixed number of lists

We prove Theorem 1. Fix $k \ge 2$ and ε. Throughout, A_k, Λ_k denote constants depending only on k. The parameters are

$$t := \lceil 16(k+1)^2/\varepsilon \rceil, \quad p := \tfrac{1}{2t}, \quad M := n^{\varepsilon/8}, \quad \delta := n^{-\varepsilon/16}, \quad \theta := n^{\varepsilon/(8(k-1))}, \quad \eta := n^{-(k+1)/t}.$$

Then $\eta^t = n^{-(k+1)}$ and $\delta^t \le n^{-(k+1)^2}$. Also $(k+1)/t \le \varepsilon/(16(k+1)) \le \varepsilon/(8(k-1))$, so $\theta/\eta \le n^{\varepsilon/(4(k-1))}$ and $M^2(\theta/\eta)^{k-1} \le n^{\varepsilon/2}$.

3.1 Overview

Conjuncts now read up to $k-1$ vectors. For a core u, the moderate links (those that fail often under $\mathcal{D}_1$) can form a sunflower $\{x_0, y_j\}$ with n petals sharing x_0. These links are not independent under $\mathcal{D}_1$, so the $e^{-\theta}$ argument of Lemma 6 fails. Yet the chosen link lies in the

sunflower with probability $\approx 1/n$, which is too large. The fix is to pluck again one level down, at the *frame* $(u, \{x_0\})$. In general a frame is a core u with a set X strictly inside some link of u. Each frame gets its own check $Q_{u,X}$, built from a maximum family of moderate links with pairwise disjoint *sub-links* $\ell \setminus X$. Frames are processed from the deepest level up. At an OR gate, the chosen vectors of a failing conjunct are confined one at a time to small hitting sets $H_{u,X}$ along a chain $\emptyset = X_0 \subsetneq X_1 \subsetneq \cdots$, which gives the error $n^{-k+o(1)}$. For $k = 3$ the only frames are $(u, \emptyset)$, and the construction is that of §2.

3.2 Distributions

$\mathcal{D}_1 = \mu_p$. Then $\Pr_{\mathcal{D}_1}[k\mathrm{Int} = 0] \le n^k(1-p^k)^d \le n^k e^{-p^k d} \le 1/n < \frac{1}{2}$ once $d \ge (k+1)\ln n/p^k$, so $c := (k+1)/p^k = (k+1)(2t)^k \le (k+1)(33(k+1)^2/\varepsilon)^k$ suffices.

$\mathcal{D}_0$: draw *chosen contents* $\gamma_1, \ldots, \gamma_k \in \{0,1\}^d$ as follows. At each coordinate independently, a uniformly random one of the k gets 0 and the others get 1. Independently, draw uniform indices $a_1, \ldots, a_k \in [n]$. Vector a_i of list i gets content γ_i, and every other vector is 1^d. Let C be the chosen k-tuple. Every sample is a 0-input.

Lemma 8. *(i) The indices are independent of the contents.*

(ii) Let S be a set of chosen vectors. Given the indices and the contents of S, any set Y of further chosen vectors with $|S| + |Y| \le k-1$ has a law on its $|Y|d$ bits that dominates $\mu_{1/2}^{\otimes |Y|}$.

Proof. (i) holds by construction. For (ii), the law is a product over coordinates, so it suffices to couple one coordinate. Where S has a zero, Y is all ones. Otherwise the zero lies uniformly among the $r := k - |S|$ chosen vectors outside S. With $m := |Y| \le r-1$, the pattern of Y is the single-zero pattern e_i with probability $1/r$ for each $i \in Y$, and 1^m with probability $(r-m)/r$. Let $U \ne \emptyset$ be an up-set and $A := \{i : e_i \in U\}$. Every $P \in U$ has its zeros inside A, since the single-zero pattern at any zero of P dominates P. Hence $\mu_{1/2}^m(U) \le 2^{|A|-m}$, while our law gives U mass $(r - m + |A|)/r \ge (1+|A|)/(m+1)$. With $b := m - |A|$ we need $2^{-b} \le 1 - b/(m+1)$ on $[0, m]$. This holds because the right side is linear and 2^{-b} is convex, with equality at $b = 0$ and $2^{-m} \le 1/(m+1)$ at $b = m$. Strassen's theorem gives a monotone coupling. □

At $|S| + |Y| = k$ the domination fails. This case is never used.

3.3 Approximators and frames

A *slot* is a set of 1 to $k-1$ vectors from distinct lists. Conjuncts are monotone and saturated, and (P) means $\Pr_{\mathcal{D}_1}[B = 0] \ge \delta$, as in §2.

Definition 9. An *approximator* is a constant, or a conjunction $F = \bigwedge_J B_J$ with one conjunct per slot and a set T of fewer than $(k-1)M$ *cores*, such that:

(B1) every conjunct satisfies (P);

(B2) every slot meets T. Each slot with $|J| \ge 2$ is assigned the core $\mathrm{core}(J)$, the vector of $J \cap T$ in the lowest-numbered list, and has *link* $\ell(J) := J \setminus \{\mathrm{core}(J)\}$. 1-slots are never assigned, switched off or dropped after step 1 of normalization;

(B3) F is the output of the normalization below.

Frames. A *frame* is a pair (u, X) with $u \in T$ and $X \subsetneq \ell(J)$ for some slot J assigned to u. Its *level* is $|X| \in \{0, \ldots, k-3\}$, and its *frame slot* is $K(u, X) := \{u\} \cup X$, of size $\le k-2$. A slot J

is *in* (u, X) if $\mathrm{core}(J) = u$ and $X \subsetneq \ell(J)$. Its *sub-link* is $\ell(J) \setminus X$, which is nonempty and has size $\le k - 2 - |X|$. Given conjuncts B, a slot J in (u, X), and values (w, v_X) of u and X, let

$$\beta_J^X(w, v_X) := \Pr_{\text{sub-link}\sim\mu_p}[B_J(w, v_X, \cdot) = 0].$$

This is nonincreasing in (w, v_X). Call J *moderate* at (w, v_X) if $\beta_J^X > \eta$. Define:

- $\nu_{u,X}(w, v_X)$: the maximum number of moderate slots in (u, X) with pairwise disjoint sub-links;
- $Q_{u,X} := [\nu_{u,X} < \theta/\eta]$. It is monotone because ν is nonincreasing, and saturated because a 1^d argument makes every β zero;
- $H_{u,X}(w, v_X)$: the union of the sub-links of the maximal disjoint family of moderate slots chosen greedily in a fixed order of the slots. It meets every moderate sub-link, and $|H_{u,X}| < (k - 2 - |X|)\theta/\eta$ whenever $Q_{u,X}(w, v_X) = 1$.

Determinism. Every choice in the construction (the disjoint family in step 2 below, the cores, the families defining $H_{u,X}$, and the anchors in Lemma 13) is made by a fixed deterministic rule. So every set and function below is a deterministic function of the circuit and of its stated arguments, which the union bounds rely on.

3.4 Normalization

$N(G)$, for a conjunction G of saturated conjuncts with at most $A_k M n^{r-1}$ slots of each size r:

1. Drop the conjuncts violating (P).
2. Take the maximal family of pairwise disjoint slots chosen greedily in a fixed order. If it has $\ge M$ members, output 0. Otherwise let T be its union and assign cores.
3. For the stages $s = k - 3, k - 4, \ldots, 0$ (deepest first): let $B^{(s)}$ be the conjuncts at the start of stage s. The frames of stage s are the level-s frames determined by the slots present at that moment, i.e. the G-slots surviving step 1 and the slots created at deeper stages. For every such frame (u, X), with all quantities computed from $B^{(s)}$:
   (a) *switch-off*: for every slot J in (u, X), replace B_J by $B_J \vee [\beta_J^X(w, v_X) \le \eta]$;
   (b) *enforce*: if $\Pr_{\mu_p}[Q_{u,X} = 0] \ge \delta$, AND $Q_{u,X}$ into the conjunct on $K(u, X)$. If that conjunct is absent, including when it was dropped in step 1, create it as $Q_{u,X}$ with the fixed core rule. Otherwise $Q_{u,X}$ is *virtual*.
4. Drop the slots of size ≥ 2 that now violate (P).

Gates are handled as in §2. A leaf is an input bit, bit j of some vector v. Its approximator is the 1-slot conjunct on $\{v\}$ that reads this bit, with $T = \{v\}$; it satisfies (P) since $1 - p \ge \delta$. For OR, $G := \bigwedge(B_J \vee B'_{J'})$ over the pairs for which $J \cup J'$ is a slot. Omitting the other pairs only raises G, so $G \ge F_g \vee F_h$. Pairs that repeat a list are $\equiv 1$ on $\mathcal{D}_0$; those with $|J \cup J'| = k$ are charged in Lemma 13.

Lemma 10 (normal form)**.** *Every enforcement on a slot J happens at stage $|J| - 1$, and every switch-off on J at a stage $\le |J| - 2$. Hence the final conjunct is*

$$\mathrm{final}_J = \Big(B_J \wedge \bigwedge \textit{enforced } Q\textit{'s on } J\Big) \vee \bigvee \textit{indicators on } J,$$

with $B_J := 1$ for a created slot. A slot created at stage s joins only frames of level $< s$, and no stage-s frame gains members after stage s.

Proof. An enforcement on J needs $J = K(u, X)$ with $|X| = |J| - 1$. A switch-off needs $X \subsetneq \ell(J)$, so $|X| \le |J| - 2$. A created slot K has $|\ell(K)| = s$. □

Invariant (I). If the final conjunct of a slot J with core u is 0 at some values, then for every $X \subsetneq \ell(J)$ the stage-$|X|$ indicator is 0 there. So J is moderate in (u, X) with respect to $B^{(|X|)}$, the family that defines $\nu_{u,X}$ and $H_{u,X}$. This follows from Lemma 10.

Sizes. A normalized approximator has at most $|T|\binom{k-1}{r-1}n^{r-1}$ slots of size r, created slots included. AND at most doubles this. At OR, each slot of G contains either a common core, or a core of T_g and a different one of T_h. So G has at most $(k-1)M\binom{k-1}{r-1}n^{r-1}+(k-1)^2M^2\binom{k-2}{r-2}n^{r-2} \le A_kMn^{r-1}$ slots of size r, for $M \le n$ and $A_k := 2k^2 2^k$.

3.5 Error analysis

Lemma 11 (normalization)**.** $\Pr_{\mathcal{D}_0}[N(G) = 1, G = 0] \le \Lambda_k M(\delta^t + \eta^t)/n$ *and* $\Pr_{\mathcal{D}_1}[N(G) = 0, G = 1] \le (1-\delta)^M + \Lambda_k M n^{k-3} e^{-\theta}$.

Proof. $\mathcal{D}_0$. If $G = 0$, some G-slot $J \subseteq C$ has $B_J = 0$ at the chosen values, which has probability $n^{-|J|}$, and $N(G) = 1$ needs the final conjunct on J to be 1. If J was not dropped and no indicator on J fires at the chosen values, then by Lemma 10 the final conjunct is $\le B_J = 0$ there. If an indicator fires, let s^* be the deepest stage at which one fires, in frame (u, X). By Lemma 10, $B_J^{(s^*)} = 0$ at the chosen values while $\beta_J^X \le \eta$. Given the contents of $\{u\} \cup X$, the sub-link dominates $\mu_{1/2}$ (Lemma 8, $|J| \le k-1$). Lemma 4 applied to the fixed monotone function $y \mapsto B_J^{(s)}(w, v_X, y)$ bounds each event $[B_J^{(s)}(\text{chosen}) = 0 \wedge \beta_J^X \le \eta]$ by η^t. There are at most 2^{k-2} such (s, X) per slot. A dropped conjunct that is 0 costs $\le \delta^t$ in the same way. Summing $n^{-r} \cdot A_k M n^{r-1}(2^{k-2}+2)$ over $r \le k-1$ gives the bound.

$\mathcal{D}_1$. Truncation costs $(1-\delta)^M$, since the disjoint members are independent. Otherwise $N(G) \ge G \wedge \bigwedge(\text{enforced } Q\text{'s})$, so suppose $G = 1$, and let (u, X) be at the deepest stage s with a failing enforced Q. Every $B^{(s)}$-conjunct is then 1. On G-slots, $B^{(s)} \ge B \wedge (\text{deeper enforced } Q\text{'s}) = 1$. Created slots are ANDs of passing Q's. But $Q_{u,X}(w, v_X) = 0$ gives $\ge \theta/\eta$ moderate slots whose sub-links are pairwise disjoint and disjoint from $\{u\} \cup X$, since $u = \text{core}(J) \notin \ell(J)$. The family is fixed by (w, v_X), and $B_J^{(s)}(w, v_X, \cdot)$ reads only its sub-link. So $\Pr[Q_{u,X} = 0 \wedge \text{all members} = 1] = \mathbb{E}[1\{Q_{u,X} = 0\}\prod_J(1-\beta_J^X)] \le (1-\eta)^{\theta/\eta} \le e^{-\theta}$. There are at most $|T|\sum_{s \le k-3}\binom{k-1}{s}n^s \le \Lambda_k M n^{k-3}$ frames. □

Lemma 12 (sequential confinement)**.** *Let $I_1, \dots, I_m$ be independent and uniform on $[n]$. For each j, let R_j be a set of size at most ρ_j that is a function of $I_1, \dots, I_{j-1}$. Then* $\Pr[I_j \in R_j \text{ for all } j] \le \prod_j \rho_j/n$.

Proof. By induction on m. Given $I_1, \dots, I_{m-1}$, the set R_m is fixed and I_m is uniform, so $\Pr[I_m \in R_m \mid I_1, \dots, I_{m-1}] \le \rho_m/n$. □

Lemma 13 (OR gate)**.** $\Pr_{\mathcal{D}_0}[G = 1, F_g = F_h = 0] \le \Lambda_k M(\delta^t + \eta^t)/n + \Lambda_k M^2(\theta/\eta)^{k-1}/n^k$.

Proof. Excluded events. For each frame (u, X) of g or h with $K := K(u, X) \subseteq C$ (probability $n^{-|K|}$, $|K| \le k-2$), exclude:

(x1) $Q_{u,X}$ is virtual and $Q_{u,X}(\text{chosen}) = 0$. By Lemma 8 with $Y = K$ and Lemma 4, this is $\le \delta^t$;

(x2) $Q_{u,X}$ is enforced, $Q_{u,X}(\text{chosen}) = 0$, and a shallower indicator on K fires at the chosen values. As in Lemma 11, take the union over the $\le 2^{|K|-1}$ fixed pairs (s, X') of the events $[B_K^{(s)}(\text{chosen}) = 0 \wedge \beta_K^{X'} \le \eta]$, conditioned on the contents of $\{\text{core}(K)\} \cup X'$. This is $\le 2^k \eta^t$;

(x3) $Q_{u,X}$ is enforced, $Q_{u,X}(\text{chosen}) = 0$, no indicator on K fires, and K was dropped in step 4. The conjunct of K is then 0 at the chosen values, so this is $\le \delta^t$.

Over at most $\Lambda_k M n^{|K|-1}$ frames of each size, these cost $\le \Lambda_k M(\delta^t + \eta^t)/n$. Work outside them, on $E := [G = 1, F_g = F_h = 0]$.

Spanning. All final conjuncts are saturated (B_J, every Q, and every indicator, since a 1^d argument makes $\beta = 0$). So a failing conjunct has its slot inside C. A *failing pair* is (J, J') with J a slot of F_g, J' a slot of F_h, and both final conjuncts 0. If $J \cup J' \subsetneq C$, it is a slot and G's conjunct on it fails. So every failing pair spans C.

Minimal pair. Fix a failing pair with $|J| + |J'|$ minimal. Each side is *terminal*: for its core u and every $X \subsetneq \ell(J)$, $Q_{u,X}(\text{chosen}) = 1$. Otherwise, by (x1) that Q is enforced on $K(u, X) \subsetneq J$. By (x2) and (x3) the final conjunct on K exists and is 0, so (K, J') is a smaller failing pair. For $X = \emptyset$, $K = \{u\}$ is a 1-slot, which is never switched or dropped. A 1-slot side $\{v\}$ has $v \in T$; we treat v as its core, with an empty chain.

Chains. Let J be a terminal side of g with core u and content w. Set $X_0 := \emptyset$. While $X_i \subsetneq \ell(J)$, J is moderate in (u, X_i) by (I), so its sub-link meets $H^g_{u,X_i}(w, v_{X_i})$ in an *anchor* h_{i+1}; set $X_{i+1} := X_i \cup \{h_{i+1}\}$. The chain enumerates $\ell(J)$. Define the confining set

$$R_{i+1} := \begin{cases} H^g_{u,X_i}(w, v_{X_i}) & \text{if } Q_{u,X_i}(w, v_{X_i}) = 1, \\ \emptyset & \text{otherwise.} \end{cases}$$

It has size $< (k-2)\theta/\eta$ on every sample. It is a function of the identities and contents of $u, h_1, \dots, h_i$, and $h_{i+1} \in R_{i+1}$ on the event, by terminality.

Counting. Condition on all chosen contents. By Lemma 8(i), the indices of the k lists are then independent and uniform. The *shape* of a side is its core list, its link length and the ordered lists of its anchors. It takes at most $k \sum_{m=0}^{k-2} (k-1)!/(k-1-m)! \le e \cdot k!$ values. A pair σ of shapes fixes an order of the lists: the core of J, then J's chain, then the core of J' if it is new, then $J' \setminus J$ in the order of J''s chain. Let E_σ be the event that each index, in this order, lies in its *confining set*: T_g or T_h (restricted to the list) for a core, and R_{i+1} for an anchor. Elements that appear again are not confined again. By the minimal-pair and chain arguments, $E \setminus \mathcal{X} \subseteq \bigcup_\sigma E_\sigma$. Each confining set is a function of the contents and of the indices earlier in the order, because a predecessor in J''s chain lies in J or comes earlier in that chain. So Lemma 12 applies: a core contributes $|T|/n < (k-1)M/n$, and each confined non-core element $< (k-2)\theta/(\eta n)$. If the cores coincide, $k-1$ non-core elements are confined; otherwise $k-2$. Hence

$$\Pr[E \setminus \mathcal{X}] \le (e \cdot k!)^2 \Big[\tfrac{(k-1)M}{n} \Big(\tfrac{(k-2)\theta}{\eta n} \Big)^{k-1} + \Big(\tfrac{(k-1)M}{n} \Big)^2 \Big(\tfrac{(k-2)\theta}{\eta n} \Big)^{k-2} \Big] \le \Lambda_k M^2 (\theta/\eta)^{k-1}/n^k. \quad \square$$

Proof of Theorem 1. Let Φ be a monotone circuit for $k\mathrm{Int}_{n,d}$. As for Theorem 2, the output approximator F satisfies $F \ge \Phi$ off the $\mathcal{D}_1$-errors and $F \le \Phi$ off the $\mathcal{D}_0$-errors. Per gate, $\varepsilon_0 \le \Lambda_k M(\delta^t + \eta^t)/n + \Lambda_k M^2 (\theta/\eta)^{k-1}/n^k \le \Lambda_k n^{-k+\varepsilon/2}$, and $\varepsilon_1 \le (1-\delta)^M + \Lambda_k M n^{k-3} e^{-\theta} \le \Lambda_k n^k \exp(-n^{\min(\varepsilon/16, \varepsilon/(8(k-1)))})$. For $k = 2$ there are no frames, so the second term is 0. If $F \equiv 0$ then $\frac{1}{2} \le |\Phi|\varepsilon_1$. Otherwise every conjunct meets T, so $\Pr_{\mathcal{D}_0}[F = 0] < (k-1)M/n \le \frac{1}{2}$ and $\frac{1}{2} \le |\Phi|\varepsilon_0$. Hence $|\Phi| \ge n^{k-\varepsilon}$ for large n. For the last sentence of the theorem, given $d = \omega(\log n)$ and $\varepsilon > 0$, eventually $d \ge c(\varepsilon) \log n$. $\square$

4 Limits of routes to refuting OVH

4.1 Block-CNF satisfiability for constant block size

This section uses its own notation, independent of §§2–3. A *block-CNF* has variables $x_\ell \in [D]$ ($\ell \le k$) and clauses that are disjunctions of block-literals $[x_\ell \in A]$ with $A \subseteq [D]$. Let $v := k\lceil \log_2 D \rceil$. k-OV with D vectors per list in dimension d is a block-CNF with d clauses, whose solutions are the orthogonal k-tuples. We assume:

(H) for every q and $\sigma > 0$, q-SAT on v variables is decidable in $O(2^{\sigma v})$ time.

By the sparsification lemma [8], (H) is equivalent to the failure of ETH. So it holds if the claimed algorithms for 3-SAT [1, 2] are correct.

Theorem 14 (conditional on (H))**.** *For fixed D, Δ and $\sigma > 0$, satisfiability of block-CNFs with at most Δv clauses can be decided in $O(2^{\sigma v})$ time. In particular, k-OV with a constant number D of vectors per list in dimension $\Delta k\lceil \log_2 D\rceil$ can be decided in $D^{o(k)}$ time as $k \to \infty$.*

Proof sketch. We may assume each clause has at most one literal per variable and each literal is consistent with the current domain R_ℓ of its variable. Fix a width w. While some clause has more than w literals, pick w of them, C', and branch: either some literal of C' holds, and the clause is replaced by C'; or every literal of C' fails, which strictly shrinks w domains. With the measure $\sum_\ell \log_2 |R_\ell|$ the number of leaves is $2^{v\lambda(w)}$, where $\lambda(w) \to 0$. At a leaf, every clause has width $\le w$. Writing each x_ℓ in binary turns the leaf into a $(w\lceil \log_2 D\rceil)$-CNF on v variables with $O(\Delta v D^w)$ clauses, together with clauses excluding values outside R_ℓ, which (H) decides in $2^{\sigma' v}$ time. Choose w and then σ'. □

Theorem 14 concerns constant D only. OVH concerns $D = n \to \infty$, where it says nothing.

4.2 Linear deformation designs (heuristic)

We have not defined a deformation design or its cost precisely, so this subsection is a heuristic, not a theorem. Consider the deformation/Taylor-interpolation framework of [1] applied to $\#\mathrm{OV} = \sum_{i,j} \prod_o (1 - \alpha_o \beta_o)$ for two lists of N vectors in dimension $d = c \log_2 N$. Consider designs with linear measurements of rank at most r and deformation degree R over Q points.

- A corank argument shows that at each point at least $d - 2r$ input variables move, hence at least $(d - 2r)/2$ coordinates.
- A determinant argument ($(T-t)^\kappa$ divides $\det \Pi(T)$, whose degree is $\le dR$) gives $Q(d-2r) \le 2dR$, which forces Taylor order $q \ge d - 2r - 1$.
- If sub-quadratic design sums need $r \le \log_3 N \approx 0.631 \log_2 N$, a threshold we have not justified rigorously, then inclusion–exclusion over coordinates enumerates $\ge 2^{(d-2r)/2-1}$ subsets of cost $\ge N$ each, in total $N^{1+(c-1.262)/2-o(1)}$. This is $N^{2+\Omega(1)}$ for $c > 2 + 2/\log_2 3 \approx 3.26$.
- If the inclusion–exclusion ranges over the $2d$ input variables, the count is $\ge 2^{d-2r-1}$, giving $N^{c-0.262-o(1)}$, which is $N^{2+\Omega(1)}$ already for $c > 1 + 2/\log_2 3 \approx 2.26$.

4.3 Equality rank of disjointness matrices

Williams [5] showed that if DISJ_k (with entries $[x \cap y = \emptyset]$) has equality rank $2^{o(k)}$, then OVH fails. The equality rank $r(k)$ is the least r with $\mathrm{DISJ}_k = \sum_{i \le r} \alpha_i E_i$, where the E_i are equality matrices $[u(i) = v(j)]$. Williams's SAT searches [5] found decompositions with all coefficients equal to 1, which give $r(k) \le k$ for $k \le 5$. Our search allows arbitrary real coefficients. For fixed r, feasibility depends only on the *type* of α (which subset sums equal 0 or 1). We enumerate the finitely many realizable types exactly and solve one SAT instance per type (`code/equality_rank_sat.py`).

Theorem 15 (computer-assisted)**.** *Over $\mathbb{R}$: $r(1) = 2$, $r(2) = 2$, $r(3) = 3$, $r(4) = 4$. For $k = 4$, $r = 3$ there are 32 types: 8 are trivially infeasible and 24 are UNSAT.*

Certificates. Explicit partitions of the ones of DISJ_k into $2, 2, 3, 4, 5$ equality matrices for $k = 1, \ldots, 5$ (certifying $r(1), r(2) \le 2$, $r(3) \le 3$, $r(4) \le 4$ and Williams's $r(5) \le 5$) are in

`data/upper_bound_certificates.log`. `code/verify_certificates.py` checks them without the search code. The lower bounds rest on the SAT runs.

Proposition 16 (computer-assisted)**.** $4 \le r(5) \le 5$*, where the upper bound is due to Williams [5]. Of the 168 types for* $r = 4$*, 20 are trivial and 147 are UNSAT. The remaining type is a pure partition. So* $r(5) = 4$ *iff the ones of* DISJ_5 *split into 4 equality matrices. This is* open*: it was not decided by several hours of CDCL search, including a cube-and-conquer run in which all 24 cubes hit a 900 s limit.*

Proposition 17. *The fractional cover number of* DISJ_k *by equality submatrices is* $7/4$*,* $23/10$*,* ≈ 2.692 *and* ≈ 3.306 *for* $k = 2, 3, 4, 5$ *(certified lower bounds* $7/4$*,* $23/10$*,* $13471/5005$*,* $8255/2499$*). Weights of either sign, valid for partitions, give the same values. So weighting arguments cannot prove* $r(4) \ge 4$ *or* $r(5) \ge 5$*, even for partitions. The computation is in* `code/fractional_cover.py`*.*

Ye's degree bound [7], combined with [6, Thm 7.2], counts OV in $n \cdot 2^{(0.3199+o(1))d} \approx n \cdot 1.248^d$ time, improving the $n \cdot 1.26^d$ of [6]. Beating it with a constant-size decomposition of DISJ_k needs $r(k) < 1.248^k$. Since DISJ_4 is a submatrix of DISJ_k, $r(k) \ge r(4) = 4$ for $k \ge 4$, so the first chances are $r(7) = 4$ and $r(8) = 5$.

5 Verification status

Result	Verification so far
Theorem 1	Full proof; 3 AI referee reads, minor fixes applied (constants, the count of shapes in Lemma 13, wording), then one further AI read. Brute-force checks in `code/general_k_checks.py`: Lemma 8 ($k \le 8$, $\lvert Y\rvert \le 5$), the reveal order of Lemma 13 ($k \le 6$), the parameters ($k \le 7$). **No human or formal check.**
Theorem 2	Full proof; 3 AI referee reads, minor fixes applied. **No human or formal check.**
Theorem 14	Proof sketch; not reviewed. Conditional on (H), i.e. on [1, 2] via [8].
§4.2	Heuristic only; not a theorem.
Theorem 15, Proposition 16	Upper bounds: explicit certificates, checked independently. Lower bounds: CaDiCaL 1.9.5, logs in `data/`; no DRAT certificates; type enumeration not independently checked.
Proposition 17	Floating-point LP; lower bounds have exact certificates, optimality does not.

6 Open problems

1. Human verification of Theorem 1, especially Lemma 13 (the reveal order) and invariant (I); ideally also a Lean formalization.
2. A check of the literature after July 2026 for monotone k-OV bounds.
3. Unbalanced OV: confirm that the argument of [3] directly gives $(nN)^{1-o(1)}$ for lists of sizes n and N.
4. The sharp monotone trade-off for two lists: is any monotone circuit subquadratic at some $c > 1$?
5. Decide $r(5)$, e.g. by lookahead cubing with DRAT proofs.

Use of AI

The author chose the problem and directed the sessions; Claude (Anthropic, Claude Opus 5.5) did the literature search, proofs, experiments, code and write-up. Review was by separate instances of the same model.

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